

A MORE BEAUTIFUL QUESTION

The Power of Inquiry to Spark Breakthrough Ideas

WARREN BERGER



The Big “So What”

Questions are an important discovery tool. In our dynamic world, the ability to ask the right questions has become increasingly important. [This book explains why it's crucial to question all aspects of our work and personal lives, and how to use inquiry to uncover creative solutions and breakthroughs.](#)



Introduction

As a journalist, Berger has widely researched and interacted with some of the world's most brilliant innovators and creators. He found that many breakthroughs can be traced back to the types of questions that are formulated and answered. [In a fast-changing world where knowledge can become obsolete quickly, the ability to ask the right questions and break old ways of thinking has become more important.](#)

Without action, questions will be philosophical at best. However, the right questions, when combined with action, bring innovation. [The book addresses “beautiful questions”, which are questions that are ambitious yet actionable.](#) These catalyze change by helping us to see and think differently, and motivate action.

[In this summary, we've condensed the key ideas into 3 parts:](#)



Why Ask Questions?



The 3-Part Framework of Innovative Questioning



Putting Inquiry to Use

KEY QUOTES

“In such times (of constant change), the ability to ask big, meaningful, beautiful questions... can be the first steps in moving beyond old habits and behaviors as we embrace the new.”

“In a time when so much of what we know is subject to revision or obsolescence, the comfortable expert must go back to being a restless learner.”

“Whether or not we were ever properly taught how to question, we can develop the skill now, on our own, in our own spaces.”

“We're all hungry for better answers. But first, we need to learn how to ask the right questions.”



Part 1: Why Ask Questions?

The Power of Questions

Today, we can find information everywhere, and knowledge has become a form of commodity.



Many people are feeling overwhelmed by the rapid rate of change, the sheer amount of information, and the wide range of data sources—some of which can be inaccurate or conflicting. Thus, having more data can be less useful than having the ability to contextualize and look at things from different angles, so you can decide what to believe.



Even technology is better at helping us to document and organize information than to help us ask good questions. Berger argues that we should no longer focus on teaching students to memorize facts, which is better done by computers. Instead, we should train people to (a) ask the right questions, and (b) frame the questions in a way that can guide computers to extract the best possible responses from the vast amount of information now available.

Questions are a powerful way to achieve creativity, innovation and change.



Questions frame our thoughts, and can be used to expand/focus our thinking:

- Asking open or speculative questions (e.g. “why is punishment effective?”) can help us to expand our thinking. On the other hand, asking closed, convergent questions (e.g. “is punishment effective?”) can help us to direct or focus our thinking, to find consensus or specific answers.
- The tone of our questions also affects the outcome. For example, a question like, “How can we possibly survive this?” can have a very different effect from, “How can we make the most of this situation?”



The right questions, when followed through with action, can lead to innovation and change. For example:

KEY QUOTES

“The value of questions is going to keep rising as that of answers keeps falling.”

“In some sense, innovation means trying to find and formulate new questions that can, over time, be answered.”

- The question, “What if we could paint over our mistakes?”, led to the invention of the “Liquid Paper” correction fluid in the 1950s.
- The question, “Why aren’t the players urinating more after games?” led to the creation of the sports drink, Gatorade.

Why do we stop questioning?

Research shows that children ask about 40,000 questions between the ages of 2 to 5. Each time the brain connects stimuli or thoughts, synapses are formed, contributing to rapid brain development during this period. As students move through pre-school to elementary school and high school, the number of questions they ask decrease drastically.

If questioning is so important, why don’t more of us do it? Why isn’t questioning an official part of education and training? Berger suggests several possible reasons:



In most schools, the teachers are under pressure to impart tons of mandated information, and there’s no time for extra questions.

- When children are fed information and offered answers to questions they haven’t yet asked, it dampens their curiosity.
- Students are also discouraged from asking questions, and learn quickly that there are more “right” answers from text books or Google than from asking people around them. Such learning habits continue into the workplace.



Questions seem to be disruptive and challenge authority and status quo. Leaders, authority and experts are expected to have the knowledge and answers, and questioning brings a sense of uncertainty and vulnerability.



It’s not easy to find answers to meaningful questions, and many people don’t have the time or patience for them. At the workplace, people are also too busy trying to take action, and asking questions seems unproductive.

KEY QUOTES

“When we start teaching too much...we’re inadvertently cutting off paths of inquiry and exploration that kids might otherwise pursue on their own.”

Can education be built on questions?

“Teaching to the test” may have been effective in producing labour for routine factory and office work in the past. However, **in our modern society, what would be the right model of education?** How can we nurture learners who can be productive and thrive in the 21st century?

When students ask questions, they have more ownership of the question and are more engaged in finding the answers.



In 1974, Deborah Meier, a renowned figure in modern education, started the alternative Central Park East school, which has delivered great results. The school helped kids to understand and apply knowledge, by focusing on **5 learning skills** or “habits of mind”: *evidence* (how you know if something is true or false), *perspective* (what are the possible viewpoints), *connection* (what are the patterns and applications), *conjecture* (what if we had different assumptions), and *significance* (why does this matter).



Montessori schools—which are **inquiry-based schools** that focus more on projects and learning rather than tests—have also nurtured many high-profile entrepreneurs, such as Wikipedia’s Jimmy Wales, Amazon’s Jeff Bezos, and Google’s Sergey Brin and Larry Page.

However, **simply asking more questions isn’t enough; people must also know how to analyze their own questions and decide which ones to pursue in depth.** Berger observed that there’s a process of inquiry that can bring about innovation and change.



The 3-Part Framework of Innovative Questioning

Problem-solving involves different phases. Each phase comes with **unique challenges, and requires a different mentality and types of questions.** By giving structure to our questions and problem-solving process, we can use the right tools and approach at each stage.

In particular, there are 3 key parts to innovative inquiry:

KEY QUOTES

“Schools in many industrialized nations were not, for the most part, designed to produce innovative thinkers or questioners—their primary purpose was to produce workers.”

“Being willing to question is one thing; questioning well and effectively is another.”



Why: Someone comes across a less-than-ideal situation, and asks “Why” to understand the situation/problem;



What If: The questioner begins to imagine possible solutions or improvements, by asking “What If”; and



How: The person zooms in on one of the possibilities, and figures out “How” to make it a reality.

In 1976, Van Phillips lost his limb in a water-skiing accident, and was given a fake foot which was no more than a block of wood with foam rubber attached to an aluminium tube. He started to question why prosthetic feet weren’t better designed, and went on to study prosthetics to learn more about the field. After 10 years of experimenting with different prototypes, he finally developed the FlexFoot line of prosthetic foot products, with breakthrough designs that changed how amputees could move and live.

Asking WHY: Step Back to Move Forward

Asking good “Why” questions help us to see and understand issues around us. This involves several components:



Step back, in at least 2 ways:

- Stop “doing”. Most of us feel pressured to keep “doing”, due to our packed schedules and endless to-do-lists. In the process, we push aside questions, as they seem to hold us back. Distance yourself from the daily busyness to consider new perspectives. Ask questions like, “why am I doing this?” or “why am I doing it this way?”
- Stop “knowing”. The more knowledge we acquire in a field, the less receptive we become to new ideas and possibilities. Step back to review your assumptions and “expert knowledge”



Take notice of what others may have missed, and **challenge assumptions**:

- Adopt the Zen principle of Soshin, or “beginner’s mind”. Bring back your childlike abilities to see things as they are, and be open to all possibilities.

KEY QUOTES

“The mind, if preoccupied with a problem or question long enough, will tend to come up with possibilities that might eventually lead to answers.”

“Asking Why can be the first step to bringing about change in almost any context.”

“With more distance, a bigger picture comes to view.”

“At least temporarily, it’s necessary to stop doing and stop knowing in order.”

“Beginner’s mind is akin to adopting a more childlike mind-set.”

- Don't be afraid to ask naïve questions, as it forces people to explain things simply and clearly.
- You can never truly be really at a dead end. If you look at things from another angle, there'll always be new details and possibilities.



Build a deeper understanding of the problem or situation through contextual inquiry. If you're too far from the problem, you cannot effectively ask the right questions. Get close to the question, so you can personally observe and understand the issue in context.



Question your questions. Sometimes, our biases and assumptions are hidden in the question we ask. Ask yourself, "What are the assumptions behind this question?" and "Should I be asking a different question?"



Use these tips to improve and deepen your questioning:

- To avoid circuitous questioning, use the 5 Whys method by Toyota founder, Sakichi Toyoda. Basically, you ask "why" 5 times:
Why am I exercising? (To lose weight);
Why am I trying to lose weight (To be healthier);
Why am I trying to be healthier? (To fall sick less);
Why am I trying to fall sick less? (To be more energetic)
Why do I want to be more energetic? (To be free to do what I want).
- The Right Question Institute (RQI) found that you can open and close a question to improve it. For example, you may start with an open question like, "Why is my boss hard to get along with?". Then, ask a closed question, "Is my boss hard to get along with?". You may realize that other colleagues do get along with your boss, and hence revise your open question to, "Why do I find it hard to get along with my boss?"



Take ownership of your chosen question, and be prepared to persist to find your answers and solutions. One subset of "Why" questions are "challenger questions". They come with an attitude of rebelliousness and an unwillingness to conform, e.g. "Why should we accept this?" or "Why can't something be done?". Such questions can be uncomfortable as they're

KEY QUOTES

"You have to be adventurous enough (and humble enough) to enter the 'know nothing' zone of a constant questioner."

"Contextual inquiry requires a commitment to the question you are exploring."

"A game-changing question can come from anyone... nonexperts or outsiders are often better at questioning than the experts."

"To question well, you must have the ability to say, 'It doesn't have to be that way.'"

disruptive and challenge authority. You're also likely to be counter-challenged, as people question, "What makes you think you know more than the experts?"

Asking WHAT IF: Imagining Possibilities

During the "What If" stage, we are in brainstorm/dreamer mode, where nothing's impossible. At this phase, it's important to allow the ideas to surface, however crazy and improbable they may seem.



Connective inquiry. Breakthrough ideas often involve mixing and matching different ideas or things that may seem totally unrelated.

- Our brains are made of huge networks of neurons. Whenever we make associations or connections, new synapses are created. Regular questioning helps to enrich the network of neural pathways, to build the foundation for insights and breakthrough ideas later.
- Although you start with a beginner's mind, make sure you acquire knowledge from different viewpoints along the way. Ask yourself questions like, "What if we combine A and B, or F and Z?"
- For example, Van Phillips did a lot of connective inquiry in exploring options for the prosthetics foot. One of the biggest breakthroughs was the Cheetah, which combined inspirations from animal legs, diving boards and curved Chinese swords, in a way that allows an amputee to walk, run and jump.



Allow ideas to incubate. During the "What If" stage, it's important to allow an idea to "sit" with you for a while, i.e. think deeply about something, then step away and allow your brain the chance to work on it and develop fresh insights. Our brains are more creative when we are relaxed. Hence, it's most productive to do such inquiry when you're having a walk, or just before bedtime (so your unconscious mind can work on the question while you sleep).



Jumbling your brain. Sometimes, our conscious mind may be resistant to certain ideas. You can "force" your brain to become more open, by deliberately coming up with ideas that are

KEY QUOTES

"Getting to a better alternative may be a long process, but it has to start somewhere—and that starting point often involves questioning the status quo."

"The What If stage is the blue-sky moment of questioning, when anything is possible."

"What If possibilities are powerful things; they are the seeds of innovation."

"Coming up with original ideas or insights...often involves the ability to combine ideas and influences, to mix and remix things that might not ordinarily go together."

"Exploring What If possibilities is a wide-open, fun stage of questioning and should not be rushed."

clearly nonsensical or “wrong”. These trigger new synapses in the brain to loosen old, persistent neural connections and make it easier to forge new ones.

- Create unusual word-combinations to get your creative juices flowing. Consider questions like, “What if we sold socks that don’t match?”, or “What if students go to school to teach rather than learn?”
- The goal of these exercises is not to find the creative ideas per se, but to train your “What If” muscles.

Asking HOW: Making Things Happen



This stage is about finding workable solutions and implementing them. It’s the slowest stage fraught with difficulties, and most of your “What If” ideas will fail in this phase.



Decide which ideas you’re willing to take to the next level. To commit to an idea, you need to give form to it, e.g. by sketching it on paper, or creating a video or prototype that you can test/get feedback.



The key to success is to fail forward: keep experimenting, learning from your failures and finding ways to improve.

- In an experiment, kindergarten kids beat Harvard MBA graduates in a challenge to build the tallest possible structure with a range of materials. The kids won because they moved quickly from “What If” to “How”, trying various combinations to learn what worked, while the adults spent too long in the “What If” phase.
- To stay positive in face of failures, focus on questions like “What have I learned?” and “How can I try a better approach?”. So long as you are failing for a different reason each time, and trying new things, you’re learning.



Collaborative inquiry. If you don’t know the solution, link up with people who may know the answers. By sharing your question or challenge, you invite like-minded people to find solutions with you. These answers are often not conclusive, but will lead you to another level of questions and exploration.

KEY QUOTES

“The How stage of questioning is...the point which things come together and then, more often than not, fall apart, repeatedly.”

“The basic principles of the test-and-learn approach apply in almost any situation where people are trying to solve problems in dynamic, uncertain conditions.”

“The trick is to go from one failure to another, with no loss of enthusiasm.”
- Winston Churchill

The point is to keep moving your idea forward, bit by bit, with continuous question-answer cycles.



Putting Inquiry to Use

Questioning in Business and at the Workplace



Most companies aren't built for questioning. Senior management tend to see questions as a threat to their credibility, chain of command and accepted practices. People also feel they have no time to ask questions, and they may not even know what to ask.



In fact, many business leaders probably rose to their positions because they could provide answers, and asking questions make them feel exposed and vulnerable. A **good leader** must also know when to stop questioning, and to start narrowing down potential outcomes to move forward.



Yet, no leader can know the answer to every question. Leaders can **empower people** to ask questions and share perspectives, to get questions and ideas from both inside and outside the company.

To succeed, **businesses must now ask more expansive questions** involving “why”, “what if”, and “how”, rather than old/closed questions involving “how many”, “how much” or “how fast”.



A good place to start is to ask **questions about purpose**, as they guide leaders to take the right actions, especially in times of uncertainty or change. Most great companies exist to answer a question and solve a problem. Good questions can refresh that sense of purpose, e.g. “Why are we in business?”, “What business are we really in?”, “If our company didn't exist, who would miss us?”, “Who were we when we were at our historical best?”, or “Who must we become to fulfil our purpose?”.



You can use **What if questions** to unlock good ideas or manage constraints, e.g. “What if we had unlimited budget to create our solutions?”, or “What if we could charge \$1 for something worth \$10?”

KEY QUOTES

“Inquiring minds can identify new opportunities and fresh possibilities before competitors become aware of them.”

“While the leader can look outside for help with questioning, certain core leadership questions can only be answered by looking inward.”

“Purpose questions are important because if you can answer them, that frees up company leaders to pursue all kinds of far-reaching opportunities and questions, knowing all the while they are on firm footing.”



Use questions to unleash your **entrepreneurial spirit**, e.g. “How can we experiment better?”, “What can we learn?” and “If we’re a startup again, what will we experiment with?”



Group brainstorming may not always deliver fresh ideas, due to the effects of peer pressure and influence. An alternative is **questionstorming**, suggested by Hal Gregersen. Unlike brainstorming (which focuses on the number of ideas), question-storming focuses on quickly improving the questions, to eventually focus on, say, the 3 best questions to be explored further.



Rather than state your mission statement as if you have already arrived, it may be more effective to list it as a **mission question**, by using the words, “*How might we?*”. For example, instead of “To inspire healthier communities through real food”, your mission question can be, “How might we inspire healthier communities through real food?”

- Presenting your mission to answer an ambitious statement can actually sound more humble and impressive. It also reminds people to keep asking if the mission is still relevant as a driving force, and if the company still lives up to its mission.
- Pair your mission question with questions like, “Why are we doing what we’re doing?” and “How might we do it better?”.

A culture of inquiry—where employees are encouraged to question—can bring dynamism and creativity to a business. At Google, anyone can ask questions, and an intra-social network is used to manage the Q&A.



However, not all companies are ready for such a culture, as it requires **system-wide changes** to policies, practices and leadership. The shift must *start at the top*, with leaders being willing to ask, accept and answer questions—including tough questions—from anywhere in the organization.



This must also be supported by a **conductive learning environment**:

- This may include **opportunities** to learn from both insiders and outsiders, collaborate with others, share work-in-progress, exchange questions and ideas. Google allows

KEY QUOTES

“When a business culture is inquisitive, the questioning, learning, and sharing of information becomes contagious—and gives people permission to explore new ideas across boundaries and silos.”

employees to devote 20% of their time to independent projects, and created Google University to make the workplace into a learning place.

- Eric Ries (the author of “Lean Startup”) suggests that both successful and “failed” experiments should be rewarded, so long as they provide valuable learning.



To grow and maintain a culture of inquiry, it also helps to add new people who're naturally inquisitive. When interviewing new hires, you can require all candidates to come with a few “Why”, “What If” and “How” questions. You can also ask questions during the interview and ask the candidates to build on these questions.

Questioning for Life

Most of us are too busy to really stop and ask why we're doing what we're doing. Berger suggests that we may even avoid the difficult questions as we silently fear the answers they'd bring.



Seminars, self-help books etc. may provide off-the-shelf answers, but they cannot replace the role of self-questioning. Only you can find the answers to your life and unique situation.

- Don't just focus on getting things done. Step back, slow down and gain clarity, so you can move forward with focus and energy.
- Rather than ask questions like “How can I get promoted?”, try asking questions like, “What's at the top of the ladder?”, “What will I do after I get there?”, “Am I enjoying the climb?”, “What am I leaving behind for the climb?”



When innovating, don't just look for what's missing; consider what is already there, e.g. existing strengths and assets. Use appreciative enquiry to make positive changes in your life, e.g. “What has worked for me before, and how can I create more of that in my life?”, “What am I naturally drawn to repeatedly?”



“Live your questions” by trying them out to see the outcome. For example, to find the best career shift, try out different projects or contract roles, to experience what really works for you. You may be surprised at what you'd discover.

KEY QUOTES

“Company leaders and managers may be able to provide some basic tips on what to look out for, but the most important thing they can offer employees is the freedom to venture outside the bubble and do their own investigations.”

“One of the hallmarks of innovative problem solvers is that they are willing to raise questions without having any idea of what the answer might be.”

“Experimentation can be thought of as, simply, the ways you act upon questions.”



Failure is inevitable in innovation and entrepreneurship. It's often recommended that you ask, "What would I do if I could not fail?" to free yourself to think big. However, success requires the ability to persist in face of failure. Alternative questions to ask could be, "How will I recover if I fail?" or "What's worth doing regardless of whether I succeed or fail?"



Questions, when properly formulated, can encourage people to break away from their own points of view, to consider different perspectives and engage in a real dialogue.

- For example, people may react defensively to statements like "Gun owners must be more responsible", but be more open to questions like, "Do you believe in gun responsibility?"
- To open up your own mind, ask questions like, "Why might they see the issue differently?", or "What are the assumptions that we're each holding?"
- Empathic questioning also helps us to narrow the gaps, connect with others, and find common ground. To strengthen relations, try questions like, "Why do I feel as if we're not getting along and how can I change that?" or "What went well in the family this week and how can we do better?"



Finding your beautiful question takes time.

- Reflect on it, and allow the ideas, possibilities and connections to sit with you. If the question is too big and overwhelming, break it down into smaller Why, What If and How questions.
- Express your personal purpose in the form of a question, as it helps you to focus, become more open to new options, and pursue your purpose adventurously without feeling the need to know all the answers.
- Test out the questions with small actions, starting with the one that resonates with you most. As you pursue a question, you may be surprised at how much you know, as well as how much you can learn.

KEY QUOTES

"There is a way back from almost anything, and once you acknowledge that, you can proceed with more confidence."

"Questions—if worded sensitively—can show respect to both sides of an issue, invite participation, and open up conversation."

"What will make it a beautiful question for you, and one worth staying with, is the passion you feel for it."

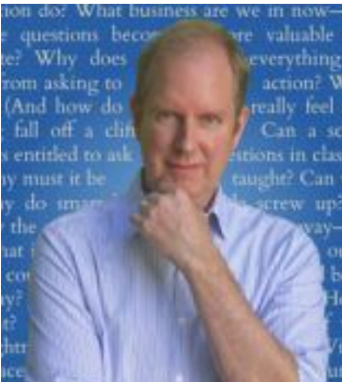
"You may have lists of things to do, goals to achieve... but if you have one compelling question, it's harder to set aside and ignore."



Other Details in the Book to Look out For

The book is organized into 5 parts. It includes over 40 segments, each structured around a question about education, innovation, problemsolving, or business, e.g. “Why do kids ask so many questions?” and “Why are you evading inquiry?”. It presents examples, background and perspectives to help us think about each question. In this summary, we’ve captured some of the key ideas and tips. You can get more articles, quizzes and resources at <http://amorebeautifulquestion.com>.

About the Author



Warren Berger (born 1958) is an American journalist, blogger and author, who has written primarily on innovation, design, mass media, and popular culture. He worked as a newspaper journalist and a magazine editor before founding his own independent writing business in 1990. Berger graduated from Syracuse University’s S. I. Newhouse School

of Public Communications.

KEY QUOTES